Exact Geometric Tails and q-Pochhammer Extraction

Finite-sinc spline approximants to Rvachev's up-function at dyadic points

Prepared for Vladimir Reshetnikov
Companion to the `ProveIt/Analysis/FabiusFunction` development

2 September 2026

Abstract

Let

$$\widehat{\mathrm{up}_n}(t) = \prod_{k=0}^{n} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right), \qquad \operatorname{sinc} z = \frac{\sin z}{z},$$

and let up be the inverse transform of the corresponding infinite product. For a fixed dyadic rational $x \in [-1, 1]$, put $q_n = \mathrm{up}_n(x)$. This article proves that the convergence of q_n is eventually not merely asymptotic: it is an exact finite sum of geometric progressions. If m is the least nonnegative integer for which $2^m x \in \mathbb{Z}$, and $d = \lfloor m/2 \rfloor$, then, for every $n \geq m$,

$$q_n = \sum_{j=0}^{d} (-1)^j a_j 4^{-j(n+1)} \mathrm{up}^{(2j)}(x) = \mathrm{up}(x) + \sum_{j=1}^{d} C_j(x) 4^{-jn}.$$

The coefficients $a_j \in \mathbb{Q}_{>0}$ are the Bell–Bernoulli coefficients of $1/\widehat{\mathrm{up}}$. The result is exact, with no hidden remainder. A one-stage sharpening covers $n = m - 1$ when m is odd, with the canonical symmetric rectangle convention in the exceptional case $m = 1$.

The proof has three ingredients. First, up_n is a compactly supported box spline with an exact truncated-power formula whose signs are the signed Thue–Morse sequence. Second, Prouhet cancellation lowers the degree of the single spline cell centered at a dyadic point from n to at most m. Third, the omitted random tail is a scaled independent copy of the full up-distribution. On the finite-dimensional space of that local polynomial, the usual all-orders deconvolution series is a polynomial in a nilpotent differentiation operator and therefore terminates exactly.

Writing $\rho = 1/4$, the constant mode $\mathrm{up}(x)$ can consequently be recovered from only $d + 1$ consecutive rational values. For any $N \geq m$,

$$\boxed{\mathrm{up}(x) = \frac{1}{(\rho;\rho)_d} \sum_{k=0}^{d} (-1)^k \rho^{k(k+1)/2} \begin{bmatrix} d \\ k \end{bmatrix}_{\rho} q_{N+d-k}.}$$

This is the first row of the inverse geometric Vandermonde system, written in finite q-Pochhammer and Gaussian-binomial form. Explicit formulas recover all geometric coefficients and all even derivatives in the terminating jet. The extraction is numerically well conditioned: the sum of the absolute weights is less than 1.969261, independently of d. Exact examples and a standard-library Python implementation accompany the article.

Contents

1 Introduction and statement of the result

The Fourier transform of Rvachev's normalized up-function is the dyadic sinc product

$$\Phi(t) := \widehat{\mathrm{up}}(t) = \prod_{k=0}^{\infty} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right). \tag{1}$$

Truncating this product after the factor with index n gives

$$\Phi_n(t) := \prod_{k=0}^{n} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right), \qquad \mathrm{up}_n := \mathcal{F}^{-1}\Phi_n. \tag{2}$$

The function up_n is a degree-n compactly supported spline. At a rational point its value is rational, and $\mathrm{up}_n \to \mathrm{up}$ rapidly in every fixed C^r-norm.

The all-orders analysis of finite dyadic sinc products naturally produces an expansion in powers of 4^{-n}. At a general point this is an asymptotic expansion. At a dyadic point, however, the expansion becomes a finite exact identity. This distinction is the central subject of the article.

Definition 1.1 (Exact dyadic level). For an interior dyadic rational $x \in (-1, 1)$, its *exact dyadic level* $m = \ell(x)$ is the least integer $m \geq 0$ such that $2^m x \in \mathbb{Z}$. Equivalently, $x = p/2^m$ in lowest terms, with p odd when $m > 0$. We put

$$d := \lfloor m/2 \rfloor, \qquad \rho := \frac{1}{4}. \tag{3}$$

The following theorem gives both the exact eventual form and its coefficients.

Theorem 1.2 (Exact terminating deconvolution). *Let $x \in (-1, 1)$ be dyadic of exact level m, and let $d = \lfloor m/2 \rfloor$. There exist positive rational numbers a_j, independent of x and n, such that for every $n \geq m$,*

$$\boxed{\mathrm{up}_n(x) = \sum_{j=0}^{d} (-1)^j a_j 4^{-j(n+1)}\, \mathrm{up}^{(2j)}(x).} \tag{4}$$

They are characterized by

$$\frac{1}{\Phi(z)} = \sum_{j=0}^{\infty} a_j (2\pi z)^{2j} \quad \text{near } z = 0. \tag{5}$$

In particular, $a_0 = 1$, so $\mathrm{up}(x)$ is the constant term.

Writing

$$C_j(x) := (-1)^j a_j 4^{-j}\, \mathrm{up}^{(2j)}(x), \tag{6}$$

we obtain the form requested in the problem.

Corollary 1.3 (Finite geometric tail). *Under the hypotheses of theorem 1.2,*

$$\boxed{q_n := \mathrm{up}_n(x) = \mathrm{up}(x) + \sum_{j=1}^{d} C_j(x)\rho^{jn}, \qquad n \geq m.} \tag{7}$$

Thus the nonconstant terms are geometric progressions with ratios

$$\rho, \rho^2, \ldots, \rho^d = \frac{1}{4}, \frac{1}{16}, \ldots, 4^{-d}.$$

The highest coefficient $C_d(x)$ is nonzero whenever $d > 0$; hence the right-hand side is a polynomial of exact degree d in ρ^n.

The constant mode can be read off without first computing the C_j. We use the standard notation

$$(a;\rho)_r := \prod_{j=0}^{r-1}(1-a\rho^j), \qquad \begin{bmatrix} r \\ k \end{bmatrix}_\rho := \frac{(\rho;\rho)_r}{(\rho;\rho)_k(\rho;\rho)_{r-k}}. \tag{8}$$

Theorem 1.4 (Single-formula extraction)**.** *Let x, m, d be as above and choose any integer $N \geq m$. Then*

$$\boxed{\mathrm{up}(x) = \frac{1}{(\rho;\rho)_d}\sum_{k=0}^{d}(-1)^k \rho^{k(k+1)/2}\begin{bmatrix} d \\ k \end{bmatrix}_\rho q_{N+d-k}, \qquad \rho = \frac{1}{4}.} \tag{9}$$

Equivalently,

$$\mathrm{up}(x) = \sum_{i=0}^{d} w_{d,i} q_{N+i}, \qquad w_{d,i} = (-1)^{d-i}\frac{\rho^{(d-i)(d-i+1)/2}}{(\rho;\rho)_i(\rho;\rho)_{d-i}}. \tag{10}$$

At $\rho = 1/4$, if

$$P_s := \prod_{r=1}^{s}(4^r - 1), \qquad P_0 := 1, \tag{11}$$

then the same weights have the integer-product form

$$\boxed{w_{d,i} = (-1)^{d-i}\frac{4^{i(i+1)/2}}{P_i P_{d-i}}.} \tag{12}$$

Formula (9) is not an approximation. When the input values $q_N, \ldots, q_{N+d}$ are exact rationals, it returns the exact rational value $\mathrm{up}(x)$.

Remark 1.5 (Indexing relative to the repository). Some documents in the accompanying repository write p_N for the inverse transform of the first N sinc factors, indexed by $k = 0, \ldots, N-1$. The notation of the present problem has

$$\mathrm{up}_n = p_{n+1}. \tag{13}$$

Keeping this one-step shift explicit prevents the common confusion between 4^{-n} and $4^{-(n+1)}$ in the differential formula.

2 Fourier, convolution, and probability normalizations

We use the Fourier convention

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x)e^{-2\pi ixt}\,\mathrm{d}x, \qquad f(x) = \int_{\mathbb{R}} \widehat{f}(t)e^{2\pi ixt}\,\mathrm{d}t \tag{14}$$

whenever the inversion formula applies classically, and otherwise in the usual distributional sense.

For $k \geq 0$, define the centered box density

$$b_k(x) := 2^k \mathbf{1}_{[-2^{-k-1},\,2^{-k-1}]}(x). \tag{15}$$

Its total mass is one and

$$\widehat{b}_k(t) = \frac{\sin(\pi t/2^k)}{\pi t/2^k} = \mathrm{sinc}\left(\frac{\pi t}{2^k}\right). \tag{16}$$

Consequently,

$$\mathrm{up}_n = b_0 * b_1 * \cdots * b_n. \tag{17}$$

This proves at once that up_n is nonnegative, has integral one, and is a piecewise polynomial of degree n. Its support radius is

$$A_n := \sum_{k=0}^{n} 2^{-k-1} = 1 - 2^{-n-1}, \qquad \operatorname{supp} \mathrm{up}_n = [-A_n, A_n]. \tag{18}$$

There is an equivalent probabilistic model. Let $U_0, U_1, \dots$ be independent random variables, each uniform on $[-1, 1]$, and set

$$S_n := \sum_{k=0}^{n} 2^{-k-1} U_k, \qquad S := \sum_{k=0}^{\infty} 2^{-k-1} U_k. \tag{19}$$

Then up_n is the density of S_n, while up is the density of S. Splitting the series after the n-th term gives the exact independent-copy identity

$$S \overset{d}{=} S_n + \delta_n S', \qquad \delta_n := 2^{-n-1}, \tag{20}$$

where S' is independent of S_n and has the same distribution as S. If

$$(D_\delta f)(x) := \frac{1}{\delta} f\left(\frac{x}{\delta}\right), \tag{21}$$

then (20) is the convolution identity

$$\boxed{\mathrm{up} = \mathrm{up}_n * D_{\delta_n} \mathrm{up}\,.} \tag{22}$$

On the Fourier side this is

$$\Phi(t) = \Phi_n(t)\Phi(\delta_n t), \qquad \Phi_n(t) = \frac{\Phi(t)}{\Phi(\delta_n t)}. \tag{23}$$

The second quotient is the source of the deconvolution expansion. The key point of this article is that, at a dyadic evaluation point, the corresponding operator series acts on a finite-dimensional polynomial space and becomes an exact finite sum.

3 The exact finite-spline formula

3.1 Truncated powers and Thue–Morse signs

Let

$$\tau_k := (-1)^{s_2(k)}, \tag{24}$$

where $s_2(k)$ is the number of ones in the binary expansion of k. Thus

$$(\tau_k)_{k\ge 0} = 1, -1, -1, 1, -1, 1, 1, -1, \dots.$$

We use $y_+^r = y^r$ for $y > 0$ and 0 for $y \le 0$. For $r = 0$, this convention gives the right-continuous Heaviside representative; only up_0 depends on the endpoint convention.

Proposition 3.1 (Exact truncated-power representation). *For every $n \ge 0$,*

$$\boxed{\mathrm{up}_n(x) = \frac{2^{n(n+1)/2}}{n!} \sum_{k=0}^{2^{n+1}-1} \tau_k \left(x + A_n - \frac{k}{2^n}\right)_+^n, \qquad A_n = 1 - 2^{-n-1}.} \tag{25}$$

The knots therefore lie on the uniform grid

$$-A_n + \frac{k}{2^n}, \qquad 0 \le k \le 2^{n+1} - 1. \tag{26}$$

Proof. Write each box as a normalized difference of two translated Heaviside functions. Convolving $n+1$ such differences and using the elementary identity that the $(n+1)$-fold integral of a point mass is $x_+^n/n!$, one obtains

$$\mathrm{up}_n(x) = \frac{\prod_{j=0}^n 2^j}{n!} \sum_{\varepsilon\in\{0,1\}^{n+1}} (-1)^{\varepsilon_0+\cdots+\varepsilon_n} \left(x + A_n - \sum_{j=0}^n \varepsilon_j 2^{-j}\right)_+^n . \tag{27}$$

Now put

$$k := \sum_{j=0}^n \varepsilon_j 2^{n-j}.$$

As ε ranges over the binary cube, k ranges once over $0, \dots, 2^{n+1}-1$,

$$\sum_{j=0}^n \varepsilon_j 2^{-j} = \frac{k}{2^n}, \qquad (-1)^{\sum_j \varepsilon_j} = \tau_k,$$

and $\prod_{j=0}^n 2^j = 2^{n(n+1)/2}$. Substitution gives (25). □

The scaled coordinate

$$y := 2^n(x + A_n) \tag{28}$$

places consecutive knots one unit apart. Since only terms with $k < y$ are active, theorem 3.1 is equivalently

$$\boxed{\mathrm{up}_n(x) = \frac{2^{-n(n-1)/2}}{n!} \sum_{0\le k\le \lfloor y\rfloor} \tau_k (y-k)^n,} \tag{29}$$

away from knots. Formula (29) makes both exact rationality and the local cancellation mechanism transparent.

Corollary 3.2 (Rational finite samples). *If $x \in \mathbb{Q}$, then $\mathrm{up}_n(x) \in \mathbb{Q}$ for every $n \ge 0$, subject only to the chosen representative at the two jumps of up_0.*

3.2 Prouhet cancellation

The signed Thue–Morse sequence annihilates low-degree polynomials on every complete dyadic block.

Lemma 3.3 (Prouhet–Thue–Morse cancellation). *For every integer $r \ge 1$ and every polynomial P of degree less than r,*

$$\sum_{j=0}^{2^r-1} \tau_j P(j) = 0. \tag{30}$$

Equivalently,

$$\sum_{j=0}^{2^r-1} \tau_j j^s = 0, \qquad 0 \le s < r. \tag{31}$$

Proof. The finite generating polynomial factors as

$$\sum_{j=0}^{2^r-1} \tau_j z^j = \prod_{h=0}^{r-1} (1 - z^{2^h}). \tag{32}$$

The right-hand side has a zero of order r at $z = 1$. Applying $(z\frac{\mathrm{d}}{\mathrm{d}z})^s$ and then setting $z = 1$ gives (31) for $s < r$. Linearity gives (30). □

We will also use the no-carry multiplicativity

$$\tau_{2^r\ell+j} = \tau_\ell \tau_j, \qquad 0 \le j < 2^r. \tag{33}$$

4 Dyadic cell centers and local degree collapse

Let $x \in (-1, 1)$ have exact level m, and define the odd integer

$$A := 2^m(x + 1). \tag{34}$$

For $m > 0$, writing $x = p/2^m$ in lowest terms gives $A = p + 2^m$, which is odd. The same is true for $m = 0, x = 0$, where $A = 1$.

Fix $n \geq m$ and put $r = n - m$. At the point x, the scaled cell coordinate (28) is

$$y = 2^n(x + A_n) = 2^n(x + 1) - \frac{1}{2} = A2^r - \frac{1}{2}. \tag{35}$$

Thus x is exactly the midpoint between two consecutive knots of up_n. The cell has half-width

$$\delta_n = 2^{-n-1}, \tag{36}$$

which is also the scale of the omitted random tail in (20).

Theorem 4.1 (Local degree collapse)**.** *Let x have exact dyadic level m. For every $n \geq m$, the restriction of up_n to the open cell*

$$(x - \delta_n, x + \delta_n) \tag{37}$$

is a polynomial of degree at most m, even though up_n has global spline degree n.

Proof. Write h for the local coordinate, $|h| < \delta_n$. Then $\lfloor 2^n(x + h + A_n)\rfloor = A2^r - 1$, and (29) gives, up to the nonzero constant $2^{-n(n-1)/2}/n!$,

$$\sum_{k=0}^{A2^r-1} \tau_k \left(A2^r - \frac{1}{2} + 2^n h - k\right)^n. \tag{38}$$

Split $k = 2^r\ell + j$, where $0 \leq \ell < A$ and $0 \leq j < 2^r$. By (33), the expression becomes

$$\sum_{\ell=0}^{A-1} \tau_\ell \sum_{j=0}^{2^r-1} \tau_j \left(2^r(A - \ell) - \frac{1}{2} - j + 2^n h\right)^n. \tag{39}$$

The coefficient of h^s is proportional to

$$\sum_{\ell=0}^{A-1} \tau_\ell \sum_{j=0}^{2^r-1} \tau_j \left(2^r(A - \ell) - \frac{1}{2} - j\right)^{n-s}. \tag{40}$$

If $s > m$, then

$$n - s < n - m = r.$$

For each fixed ℓ, the inner summand in (40) is a polynomial in j of degree less than r, so it vanishes by theorem 3.3. Every coefficient of h^s with $s > m$ is therefore zero. □

Remark 4.2 (What the cancellation really does). The global degree grows with n, but a fixed dyadic point organizes the active truncated powers into complete blocks of length 2^{n-m}. Each extra sinc factor creates one extra binary cancellation moment. The increase in global polynomial degree is exactly offset by the increase in Prouhet order, leaving a local degree bounded by the fixed denominator exponent m.

5 Bell–Bernoulli deconvolution coefficients

5.1 The moment-generating function

Let

$$M(s) := \int_{-1}^{1} e^{sy}\,\mathrm{up}(y)\,\mathrm{d}y = \Phi\left(\frac{is}{2\pi}\right). \tag{41}$$

Since up is even, M is an even entire function and

$$M(s) = \sum_{r=0}^{\infty} \frac{\mu_{2r}}{(2r)!} s^{2r}, \qquad \mu_{2r} := \int_{-1}^{1} y^{2r}\,\mathrm{up}(y)\,\mathrm{d}y. \tag{42}$$

The classical Bernoulli expansion

$$\log\frac{z}{\sin z} = \sum_{r=1}^{\infty} \frac{2^{2r-1}\,|B_{2r}|}{r(2r)!} z^{2r} \tag{43}$$

combined with the geometric sum over the dyadic scales yields

$$\log\frac{1}{\Phi(z)} = \sum_{r=1}^{\infty} \alpha_r (2\pi z)^{2r}, \qquad \boxed{\alpha_r = \frac{|B_{2r}|}{2r(2r)!\,(1-2^{-2r})}.} \tag{44}$$

Define a_j by

$$\frac{1}{\Phi(z)} = \exp\left(\sum_{r=1}^{\infty} \alpha_r (2\pi z)^{2r}\right) = \sum_{j=0}^{\infty} a_j (2\pi z)^{2j}. \tag{45}$$

Then every a_j is a positive rational number. In terms of complete exponential Bell polynomials,

$$\boxed{a_j = \frac{1}{j!} Y_j\big(1!\alpha_1, 2!\alpha_2, \dots, j!\alpha_j\big).} \tag{46}$$

Equivalently,

$$a_j = \sum_{k_1+2k_2+\cdots+jk_j=j} \prod_{r=1}^{j} \frac{\alpha_r^{k_r}}{k_r!}. \tag{47}$$

The most convenient exact recurrence is

$$\boxed{a_0 = 1, \qquad j a_j = \sum_{r=1}^{j} r\alpha_r a_{j-r} \quad (j \ge 1).} \tag{48}$$

The first coefficients are

$$\begin{aligned} a_0 &= 1, & a_1 &= \frac{1}{18}, & a_2 &= \frac{31}{16200}, \\ a_3 &= \frac{2347}{42865200}, & a_4 &= \frac{1904369}{1311675120000}, & a_5 &= \frac{3309760579}{88561680752160000}. \end{aligned} \tag{49}$$

Substitution of $z = is/(2\pi)$ into (45) gives the reciprocal moment series

$$\boxed{\frac{1}{M(s)} = \sum_{j=0}^{\infty} (-1)^j a_j s^{2j}.} \tag{50}$$

5.2 An exact inverse on polynomials

Let $D = \frac{\mathrm{d}}{\mathrm{d}z}$. If P is a polynomial, define

$$Q(z) := \int_{-1}^{1} P(z - \delta y)\,\mathrm{up}(y)\,\mathrm{d}y. \tag{51}$$

Taylor expansion terminates, and evenness of up gives the exact operator identity

$$Q = M(\delta D)P. \tag{52}$$

Lemma 5.1 (Finite polynomial deconvolution). *If P has degree at most m, and $Q = M(\delta D)P$, then*

$$\boxed{P = \sum_{j=0}^{\lfloor m/2 \rfloor} (-1)^j a_j \delta^{2j} Q^{(2j)}.} \tag{53}$$

More generally, for $0 \le r \le m$,

$$P^{(r)}(0) = \sum_{j=0}^{\lfloor (m-r)/2 \rfloor} (-1)^j a_j \delta^{2j} Q^{(r+2j)}(0). \tag{54}$$

Proof. On the vector space of polynomials of degree at most m, the operator D is nilpotent: $D^{m+1} = 0$. Therefore every formal power series in D is actually a finite polynomial in D, and multiplication of formal series is legitimate without any convergence question. Equations (52) and (50) give

$$P = M(\delta D)^{-1} Q = \sum_{j \ge 0} (-1)^j a_j \delta^{2j} D^{2j} Q.$$

All terms with $2j > m$ vanish, proving (53). Differentiating r times and evaluating at zero gives (54). □

The exactness in theorem 5.1 is stronger than observing that all sufficiently high coefficients of an asymptotic expansion vanish at a dyadic point. A vanishing formal tail alone would not rule out a remainder that is smaller than every power of 4^{-n}. Here there is no such issue: the inverse series is evaluated on a nilpotent operator, so it is literally a finite algebraic identity.

---

6 Proof of exact termination at dyadic points

6.1 The centered-cell case

Fix a dyadic x of level m and a stage $n \ge m$. By theorem 4.1, there is a polynomial P of degree at most m such that

$$P(z) = \mathrm{up}_n(x + z), \qquad |z| < \delta_n. \tag{55}$$

Define Q from P by (51) with $\delta = \delta_n$.

For $0 \le r \le m$, the convolution identity (22), interpreted distributionally when necessary, gives

$$\mathrm{up}^{(r)}(x) = \int_{-1}^{1} \mathrm{up}_n^{(r)}(x - \delta_n y)\,\mathrm{up}(y)\,\mathrm{d}y. \tag{56}$$

For $-1 < y < 1$, the point $x - \delta_n y$ lies strictly inside the centered cell, so $\mathrm{up}_n^{(r)} = P^{(r)}$ there. The two endpoints have measure zero. Consequently,

$$Q^{(r)}(0) = \mathrm{up}^{(r)}(x), \qquad 0 \le r \le m. \tag{57}$$

Applying theorem 5.1 at $z = 0$, and using $P(0) = \mathrm{up}_n(x)$, gives

$$\mathrm{up}_n(x) = \sum_{j=0}^{\lfloor m/2 \rfloor} (-1)^j a_j \delta_n^{2j} \,\mathrm{up}^{(2j)}(x).$$

Since $\delta_n^{2j} = 4^{-j(n+1)}$, this proves theorem 1.2.

The same proof yields a derivative version.

Corollary 6.1 (Terminating laws for finite-spline derivatives). *Let $0 \le r \le m$. For every $n \ge m$,*

$$\boxed{\mathrm{up}_n^{(r)}(x) = \sum_{j=0}^{\lfloor (m-r)/2 \rfloor} (-1)^j a_j 4^{-j(n+1)} \,\mathrm{up}^{(r+2j)}(x).} \tag{58}$$

Thus the derivative sequence also has a finite exact geometric tail, with at most $\lfloor (m-r)/2 \rfloor$ nonconstant modes.

6.2 A one-stage sharpening at odd dyadic levels

The universal centered-cell proof starts at $n = m$. When m is odd, the same identity already holds one stage earlier.

Proposition 6.2 (Odd-level onset). *Let $m = 2d + 1 \ge 3$, and let x have exact level m. Then (4) also holds at $n = m - 1 = 2d$. Hence the geometric law (7) is valid for every*

$$n \ge \nu(x) := 2 \lfloor m/2 \rfloor. \tag{59}$$

For $m = 1$, the same statement at $n = 0$ holds if the inverse transform of the first sinc factor is assigned its canonical symmetric value $1/2$ at $x = \pm 1/2$. Avoiding this endpoint convention, one may simply start at $n = m = 1$.

Proof. Put $n = m - 1 = 2d$. In this case x is a knot of the degree-n spline up_n, and the interval $[x - \delta_n, x + \delta_n]$ lies in the union of the two adjacent knot cells, meeting each in a half-cell. Let P_- and P_+ be the polynomial pieces in the local coordinate z, on the left and right respectively. Since a convolution of $n + 1$ boxes is C^{n-1}, the two pieces have the form

$$P_-(z) = R(z) + a_- z^n, \qquad P_+(z) = R(z) + a_+ z^n, \qquad \deg R \le n - 1. \tag{60}$$

Define their top-coefficient average

$$P_*(z) := R(z) + \frac{a_- + a_+}{2} z^n. \tag{61}$$

Because n is even, for every even $r \le n$, evenness of up shows that the contributions from the two half-intervals satisfy

$$\mathrm{up}^{(r)}(x) = \int_0^1 \left(P_-^{(r)}(-\delta_n y) + P_+^{(r)}(\delta_n y) \right) \mathrm{up}(y)\, \mathrm{d}y \tag{62}$$

$$= \int_{-1}^1 P_*^{(r)}(-\delta_n y)\, \mathrm{up}(y)\, \mathrm{d}y. \tag{63}$$

Moreover, continuity gives $P_*(0) = P_-(0) = P_+(0) = \mathrm{up}_n(x)$. Apply the finite polynomial inverse theorem 5.1 to P_*, whose degree is $n = 2d$. Only even derivatives occur, and (62) supplies exactly the required jet of up. The result is (4) at $n = m - 1$.

For $m = 1, n = 0$, the same argument averages the two constant pieces of a rectangle. Symmetric Fourier inversion assigns their half-sum at the jump. ∎

6.3 Vanishing derivatives and the highest geometric mode

Repeated differentiation of Rvachev's functional-differential equation

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big) \tag{64}$$

gives an exact Thue–Morse formula for every derivative.

Lemma 6.3 (Derivative dilation formula)**.** *For every integer $r \geq 0$,*

$$\boxed{\mathrm{up}^{(r)}(x) = 2^{r(r+1)/2} \sum_{k=0}^{2^r-1} \tau_k \ \mathrm{up}(2^r x + 2^r - 1 - 2k)\,.} \tag{65}$$

Proof. The case $r = 0$ is tautological, and the case $r = 1$ is (64). Differentiate the formula for order r, apply (64) to each term, and split each old index into the two new binary indices k and $k + 2^r$. The identities $\tau_{k+2^r} = -\tau_k$ and $2^{r(r+1)/2} 2^r 2 = 2^{(r+1)(r+2)/2}$ give the order-$r+1$ formula. ☐

The values needed below follow directly from the already proved terminating formula. At $x = 0$, level $m = 0$ gives $\mathrm{up}(0) = \mathrm{up}_0(0) = 1$. At $x = \pm 1/2$, level $m = 1$ gives $\mathrm{up}(\pm 1/2) = \mathrm{up}_1(\pm 1/2) = 1/2$, the last value being immediate from the convolution of the boxes of half-widths $1/2$ and $1/4$.

Proposition 6.4 (Dyadic derivative cutoff and nonzero top even jet)**.** *Let $x = p/2^m \in (-1, 1)$ have exact level m, and put $A = 2^m(x + 1)$. Then*

$$\mathrm{up}^{(r)}(x) = 0 \qquad (r > m). \tag{66}$$

If $m = 2d$ is even and positive, then

$$\mathrm{up}^{(m)}(x) = 2^{m(m+1)/2} \tau_{(A-1)/2} \neq 0. \tag{67}$$

If $m = 2d + 1$ is odd, then

$$\mathrm{up}^{(m-1)}(x) = 2^{m(m-1)/2-1} \tau_{\lfloor (A-1)/4 \rfloor} \neq 0. \tag{68}$$

Proof. For $r > m$, the number $2^r x$ is an even integer. Every argument of up in (65) is then an odd integer. The only odd integers in $[-1, 1]$ are ± 1, and $\mathrm{up}(\pm 1) = 0$; hence the whole sum vanishes.

For $r = m$, the arguments in (65) are even integers. Exactly one is zero, at $k = (A - 1)/2$, and every other argument lies outside $(-1, 1)$. Since $\mathrm{up}(0) = 1$, this proves (67) when m is even.

If m is odd, take $r = m - 1$. The arguments are half-integers separated by two. Exactly one equals $1/2$ or $-1/2$, at $k = \lfloor (A - 1)/4 \rfloor$. Since $\mathrm{up}(\pm 1/2) = 1/2$, formula (68) follows. ☐

Since $a_d > 0$, theorem 6.4 proves the nonvanishing assertion in theorem 1.3. It also gives an explicit sign and magnitude for the slowest-to-appear, fastest-decaying top mode $C_d(x)$.

7 The geometric Vandermonde system

Assume from now on that $N \geq m$, so no convention at up_0 is involved. Put

$$B_j := C_j(x)\rho^{jN}, \qquad B_0 = \mathrm{up}(x). \tag{69}$$

The $d + 1$ consecutive samples satisfy

$$q_{N+i} = \sum_{j=0}^{d} B_j \rho^{ij}, \qquad 0 \leq i \leq d. \tag{70}$$

In matrix form,

$$\begin{pmatrix} q_N \\ q_{N+1} \\ \vdots \\ q_{N+d} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \rho & \rho^2 & \cdots & \rho^d \\ 1 & \rho^2 & \rho^4 & \cdots & \rho^{2d} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \rho^d & \rho^{2d} & \cdots & \rho^{d^2} \end{pmatrix} \begin{pmatrix} B_0 \\ B_1 \\ \vdots \\ B_d \end{pmatrix}. \tag{71}$$

The determinant is

$$\prod_{0\le j<k\le d} (\rho^k - \rho^j) \neq 0, \tag{72}$$

so the coefficients are uniquely determined.

7.1 The first inverse row by Lagrange interpolation

Define the polynomial

$$f(z) := \sum_{j=0}^{d} B_j z^j. \tag{73}$$

Then $f(\rho^i) = q_{N+i}$, while $f(0) = B_0 = \mathrm{up}(x)$. Lagrange interpolation at the geometric nodes $1, \rho, \ldots, \rho^d$ therefore gives

$$\mathrm{up}(x) = \sum_{i=0}^{d} \left(\prod_{\substack{0\le \ell\le d \\ \ell\neq i}} \frac{-\rho^\ell}{\rho^i - \rho^\ell} \right) q_{N+i}. \tag{74}$$

For $\ell < i$, the corresponding product factors are $1/(1-\rho^{i-\ell})$; for $\ell > i$, they are $-\rho^{\ell-i}/(1-\rho^{\ell-i})$. Their product is exactly (10). Replacing i by $d-k$ and using the Gaussian-binomial definition (8) proves (9).

The operator form of the same argument is particularly compact. Let the forward shift act by

$$(\mathsf{E}q)_n := q_{n+1}. \tag{75}$$

Each factor $\mathsf{E} - \rho^j$ kills the geometric mode ρ^{jn}. Hence

$$\boxed{\mathrm{up}(x) = \frac{1}{(\rho;\rho)_d} \left[\prod_{j=1}^{d} (\mathsf{E} - \rho^j) q \right]_N.} \tag{76}$$

Expanding

$$\prod_{j=1}^{d} (\mathsf{E} - \rho^j) = \mathsf{E}^d (\rho \mathsf{E}^{-1}; \rho)_d \tag{77}$$

with the finite q-binomial theorem produces (9) in one line.

7.2 First extraction formulas

For small d, the formulas are

$$d = 0: \quad \mathrm{up}(x) = q_N, \tag{78}$$

$$d = 1: \quad \mathrm{up}(x) = \frac{-q_N + 4q_{N+1}}{3}, \tag{79}$$

$$d = 2: \quad \mathrm{up}(x) = \frac{q_N - 20q_{N+1} + 64q_{N+2}}{45}, \tag{80}$$

$$d = 3: \quad \mathrm{up}(x) = \frac{-q_N + 84q_{N+1} - 1344q_{N+2} + 4096q_{N+3}}{2835}, \tag{81}$$

$$d = 4: \quad \mathrm{up}(x) = \frac{q_N - 340q_{N+1} + 22848q_{N+2} - 348160q_{N+3} + 1048576q_{N+4}}{722925}. \tag{82}$$

The required number of samples grows only every second time the dyadic denominator exponent grows:

exact level m	number of modes d	samples needed	guaranteed start
0 or 1	0	1	$N \geq m$
2 or 3	1	2	$N \geq m$
4 or 5	2	3	$N \geq m$
6 or 7	3	4	$N \geq m$
8 or 9	4	5	$N \geq m$

For odd m, theorem 6.2 permits the first window to start at $N = m - 1$, apart from the harmless $m = 1$ point-value convention.

7.3 Uniform stability of the extraction

Although the weights alternate in sign, their total variation stays bounded as $d \to \infty$.

Proposition 7.1 (Lebesgue factor of the geometric extraction). *For* $0 < \rho < 1$,

$$\boxed{\Lambda_d(\rho) := \sum_{i=0}^{d} |w_{d,i}| = \frac{(-\rho;\rho)_d}{(\rho;\rho)_d}.} \tag{83}$$

At $\rho = 1/4$,

$$\Lambda_d(1/4) < \frac{(-1/4;1/4)_\infty}{(1/4;1/4)_\infty} = 1.96926035366826943 19\ldots. \tag{84}$$

Consequently, if each sample is perturbed by at most ε*, the extracted value is perturbed by less than* $1.969261\,\varepsilon$.

Proof. Put $k = d - i$ in the absolute-value sum and use (9) without its alternating signs:

$$\begin{aligned}\Lambda_d(\rho) &= \frac{1}{(\rho;\rho)_d} \sum_{k=0}^{d} \rho^{k(k+1)/2} \begin{bmatrix} d \\ k \end{bmatrix}_\rho \\ &= \frac{(-\rho;\rho)_d}{(\rho;\rho)_d},\end{aligned}$$

where the second equality is the finite q-binomial theorem applied to $(-\rho;\rho)_d$. The ratio increases to the displayed convergent infinite-product limit when $\rho = 1/4$. The perturbation bound is the triangle inequality. □

This bounded factor is a useful contrast with ordinary high-order Richardson extrapolation at nearly coincident nodes, whose weights can be enormous. The geometric nodes $1, 1/4, 1/16, \ldots$ remain well separated on the logarithmic scale.

8 Recovering every geometric coefficient and even derivative

The same sample window determines all coefficients, not only the constant one. Apply to (7) the shift polynomial that kills every mode except the s-th.

Theorem 8.1 (Mode projector)**.** *For $0 \le s \le d$ and $N \ge m$,*

$$\boxed{C_s(x) = \rho^{-sN} \frac{\left[\prod_{\substack{0\le j\le d \\ j\ne s}} (\mathsf{E} - \rho^j) q\right]_N}{\prod_{\substack{0\le j\le d \\ j\ne s}} (\rho^s - \rho^j)},} \tag{85}$$

where $C_0(x) = \mathrm{up}(x)$. The denominator has the closed form

$$\boxed{\prod_{j\ne s} (\rho^s - \rho^j) = (-1)^s \rho^{s(s-1)/2 + s(d-s)} (\rho;\rho)_s (\rho;\rho)_{d-s}.} \tag{86}$$

Proof. On the mode $C_r \rho^{rn}$, the numerator operator acts by multiplication with $\prod_{j\ne s}(\rho^r - \rho^j)$. This is zero unless $r = s$. For $r = s$, division by the displayed eigenvalue and by ρ^{sN} recovers C_s.

For $j < s$, write $\rho^s - \rho^j = -\rho^j(1-\rho^{s-j})$; for $j > s$, write $\rho^s - \rho^j = \rho^s(1-\rho^{j-s})$. Multiplication gives (86). □

For a completely expanded q-binomial formula, split the numerator into roots below and above s. The finite q-binomial theorem gives

$$\begin{aligned} &\left[\prod_{j\ne s} (\mathsf{E} - \rho^j) q\right]_N \\ &= \sum_{a=0}^{s} \sum_{b=0}^{d-s} (-1)^{a+b} \rho^{a(a-1)/2 + (s+1)b + b(b-1)/2} \begin{bmatrix} s \\ a \end{bmatrix}_\rho \begin{bmatrix} d-s \\ b \end{bmatrix}_\rho q_{N+d-a-b}. \end{aligned} \tag{87}$$

Equations (85), (86), and (87) are a closed-form inverse for the entire geometric Vandermonde system.

Finally, (6) converts geometric modes into the even derivative jet.

Corollary 8.2 (Even-jet recovery)**.** *For $1 \le s \le d$,*

$$\boxed{\mathrm{up}^{(2s)}(x) = \frac{(-1)^s 4^s}{a_s} C_s(x).} \tag{88}$$

Thus $d+1$ consecutive finite-spline values recover not only $\mathrm{up}(x)$ but every even derivative $\mathrm{up}^{(2)}, \ldots, \mathrm{up}^{(2d)}$ that can occur in the terminating formula.

9 The exact q-difference recurrence

Because q_n is a finite linear combination of the modes $1, \rho^n, \ldots, \rho^{dn}$, it satisfies a constant-coefficient recurrence.

Theorem 9.1 (Annihilating recurrence). *For every $n \geq m$,*

$$\prod_{j=0}^{d} (\mathsf{E} - \rho^j) q_n = 0. \tag{89}$$

Equivalently,

$$\boxed{\sum_{k=0}^{d+1} (-1)^k \rho^{k(k-1)/2} \begin{bmatrix} d+1 \\ k \end{bmatrix}_\rho q_{n+d+1-k} = 0.} \tag{90}$$

For odd m, the recurrence already applies to the window beginning at $n = m - 1$ under theorem 6.2.

Proof. Each factor $\mathsf{E} - \rho^j$ annihilates its corresponding mode. Expanding

$$\prod_{j=0}^{d} (\mathsf{E} - \rho^j) = \mathsf{E}^{d+1} (\mathsf{E}^{-1}; \rho)_{d+1}$$

by the finite q-binomial theorem gives (90). □

The recurrence provides a simple exact diagnostic. If rationally computed samples fail (90), either the starting stage is too early, the indexing convention is inconsistent, or an arithmetic error has occurred. In floating-point computation, the recurrence residual gives an internal estimate of numerical contamination.

10 Exact examples

All fractions in this section are produced by the truncated-power formula (25); no numerical approximation is involved.

Example 10.1 (Stationary levels $m = 0, 1$). At $x = 0$, one has $m = d = 0$, and

$$\mathrm{up}_n(0) = \mathrm{up}(0) = 1 \qquad (n \geq 0).$$

At $x = \pm 1/2$, one has $m = 1, d = 0$, and

$$\mathrm{up}_n(\pm 1/2) = \mathrm{up}(\pm 1/2) = \frac{1}{2} \qquad (n \geq 1).$$

Thus no decaying terms are present at levels zero and one.

Example 10.2 (The point $x = 1/4$). Here $m = 2, d = 1$. The first guaranteed samples are

$$q_2 = \frac{15}{16}, \qquad q_3 = \frac{179}{192}. \tag{91}$$

Formula (79) gives

$$\mathrm{up}\left(\frac{1}{4}\right) = \frac{-q_2 + 4q_3}{3} = \frac{67}{72}. \tag{92}$$

The derivative formula gives $\mathrm{up}''(1/4) = -8$, and $a_1 = 1/18$. Therefore

$$\boxed{q_n = \frac{67}{72} + \frac{1}{9} 4^{-n}, \qquad n \geq 2.} \tag{93}$$

The preceding value $q_1 = 1$ does not satisfy this law, so the guaranteed onset $n = m = 2$ is sharp in this example.

Example 10.3 (Odd level: $x = 1/8$)**.** Here $m = 3, d = 1$. The ordinary centered-cell theorem starts at $n = 3$, but theorem 6.2 moves the onset to $n = 2$. Indeed,

$$q_2 = 1, \qquad q_3 = \frac{383}{384}, \qquad q_4 = \frac{1531}{1536}. \tag{94}$$

Using the first two gives

$$\mathrm{up}\left(\frac{1}{8}\right) = \frac{-q_2 + 4q_3}{3} = \frac{287}{288}. \tag{95}$$

Since $\mathrm{up}''(1/8) = -4$,

$$\boxed{q_n = \frac{287}{288} + \frac{1}{18} 4^{-n}, \qquad n \geq 2.} \tag{96}$$

The value of q_4 follows automatically from the same law.

Example 10.4 (Two modes: $x = 1/16$)**.** Now $m = 4, d = 2$, so three samples are required:

$$q_4 = \frac{24575}{24576}, \qquad q_5 = \frac{1965959}{1966080}, \qquad q_6 = \frac{94365509}{94371840}. \tag{97}$$

The $d = 2$ extraction gives

$$\begin{aligned} \mathrm{up}\left(\frac{1}{16}\right) &= \frac{q_4 - 20q_5 + 64q_6}{45} \\ &= \frac{2073457}{2073600}. \end{aligned} \tag{98}$$

The even derivatives are

$$\mathrm{up}''\left(\frac{1}{16}\right) = -\frac{5}{9}, \qquad \mathrm{up}^{(4)}\left(\frac{1}{16}\right) = -1024. \tag{99}$$

Using $a_1 = 1/18$ and $a_2 = 31/16200$ gives the complete exact sequence

$$\boxed{q_n = \frac{2073457}{2073600} + \frac{5}{648} 4^{-n} - \frac{248}{2025} 16^{-n}, \qquad n \geq 4.} \tag{100}$$

This example displays the two distinct geometric ratios $1/4$ and $1/16$ and illustrates why ordinary one-step Richardson extrapolation is not enough at higher dyadic levels.

Example 10.5 (Reflection)**.** Every finite product and the limiting density are even. Hence

$$\mathrm{up}_n(-x) = \mathrm{up}_n(x), \qquad \mathrm{up}(-x) = \mathrm{up}(x),$$

and all formulas above apply unchanged after replacing x by $-x$. At $x = \pm 1$, every finite approximant vanishes because its support is strictly contained in $[-1, 1]$, and the smooth limiting function also vanishes. The endpoint sequence is identically zero.

11 An exact extraction algorithm

The mathematical procedure separates cleanly into two layers. The first layer computes rational finite-spline values by any exact method; the second layer applies a short, universal q-Pochhammer filter.

Algorithm 11.1 (Dyadic limit extraction)**.** Given a dyadic rational $x \in [-1, 1]$:

1. If $x = \pm 1$, return 0.

2. Reduce x to lowest terms and let m be the base-two logarithm of its denominator. Put $d = \lfloor m/2 \rfloor$.
3. Choose a starting stage $N \geq m$. The canonical choice is $N = m$. For odd $m \geq 3$, $N = m - 1$ is also valid by theorem 6.2.
4. Compute the $d + 1$ exact rational values

$$q_N, q_{N+1}, \ldots, q_{N+d}.$$

Formula (25) is a direct reference implementation.

5. Compute the weights from either (10) or (12), and return

$$\sum_{i=0}^{d} w_{d,i} q_{N+i}.$$

The returned fraction is exactly $\mathrm{up}(x)$.

11.1 Pseudocode

```
def extract_up_at_dyadic(x):
    # x is an exact reduced Fraction with power-of-two denominator.
    if abs(x) == 1:
        return Fraction(0)

    m = log2(x.denominator)        # exact integer
    d = m // 2
    rho = Fraction(1, 4)
    N = m                          # unconditional starting stage

    samples = [finite_up(N + i, x) for i in range(d + 1)]
    weights = []
    for i in range(d + 1):
        T = (d - i) * (d - i + 1) // 2
        w = (-1)**(d - i) * rho**T
        w /= q_pochhammer(rho, i) * q_pochhammer(rho, d - i)
        weights.append(w)

    return sum(w * q for w, q in zip(weights, samples))
```

11.2 Cost and validation

The direct truncated-power evaluator has 2^{n+1} potential terms at stage n, though only the active prefix is needed. It is ideal for transparent verification at modest levels. Faster production evaluators can exploit binary block recurrences, local spline propagation, or precomputed dyadic rows; none of those choices changes the extraction filter.

Once the samples are available, computing the weights and their dot product requires only $O(d)$ rational operations if successive products $(\rho;\rho)_i$ are cached. Since $d = \lfloor m/2 \rfloor$, this second layer is small even for fairly deep dyadic levels.

Two exact checks are especially useful:

1. repeat the extraction with the shifted window $q_{N+1}, \ldots, q_{N+d+1}$; the resulting rational must be identical;

2. evaluate the recurrence (90); its residual must be exactly zero.

If an upper bound $D \geq d$ is used instead of the exact d, the $(D+1)$-sample filter still returns the same constant mode, because it annihilates all modes through ρ^{Dn}. Underestimating d, by contrast, generally leaves an uncancelled geometric term.

11.3 Approximate samples and rational reconstruction

When q_n is evaluated numerically rather than rationally, apply the same weights at high precision. By theorem 7.1, the extraction amplifies uniform absolute error by less than a factor of 1.97. If a denominator bound is known from independent arithmetic theory, one may then use continued fractions or rational reconstruction on the high-precision result. Exact input remains preferable: formula (25) and the extraction weights involve rational arithmetic only.

12 Companion implementation and verification

The archive accompanying this article contains `up_dyadic_extraction.py`. It uses only the Python standard library and `fractions.Fraction`. The program implements:

- the exact truncated-power evaluator (25);
- finite q-Pochhammer symbols, Gaussian coefficients, and extraction weights;
- the Bell–Bernoulli recurrence (48);
- exact dyadic evaluation of up by (9);
- exact derivative evaluation by (65);
- exhaustive theorem checks through a user-selected binary level.

For example,

```
python up_dyadic_extraction.py --examples
python up_dyadic_extraction.py --x 1/16
python up_dyadic_extraction.py --verify-level 7
```

The bundled verification run checked 2376 exact identities at all 255 interior dyadic points of exact level at most seven. This includes sliding-window invariance of the q-Pochhammer extraction and the differential formula (4) at several consecutive stages. Every comparison was an equality of rational numbers; no tolerance or floating-point arithmetic was used.

The computation is evidence that the formulas have been transcribed and indexed correctly. The proofs in the preceding sections are independent of the experiment.

13 Conceptual consequences and extensions

13.1 A second proof of dyadic rationality

Classical arithmetic formulas already imply that $\mathrm{up}(x)$ is rational at dyadic x. The present construction gives a short independent route. Every q_{N+i} is rational by theorem 3.2; every weight in (10) is rational; hence their finite sum $\mathrm{up}(x)$ is rational. The argument is constructive and returns the value itself, not merely a rationality certificate.

13.2 Why the number of modes is half the binary level

The local polynomial degree is at most m, but the omitted tail is symmetric. Its moment-generating function is even, so the deconvolution operator contains only even powers D^{2j}. Therefore only

$$0, 2, 4, \ldots, 2\lfloor m/2 \rfloor$$

can contribute at the point. This is why the geometric degree is $d = \lfloor m/2 \rfloor$, rather than m, and why adding one binary denominator bit increases the sample count only every other time.

13.3 Universality of the q-filter

The extraction theorem is not specific to the up-function once the finite geometric law has been established. If any sequence has the form

$$s_n = L + \sum_{j=1}^{d} A_j \rho^{jn}, \tag{101}$$

then exactly the same weights recover L from any $d + 1$ consecutive samples. The up-function supplies an unusually rigid mechanism that proves (101) exactly: a self-similar omitted tail, a local polynomial, and Prouhet cancellation.

13.4 Beyond point values

Corollary 6.1 already extends the method to derivative samples. More generally, any linear functional depending only on the local polynomial jet inherits a finite geometric law by applying that functional to (53). This includes finite combinations of values and derivatives inside a fixed centered cell. The same algebra can therefore serve as an exact extrapolation layer for local approximation schemes built from the finite dyadic box splines.

13.5 Relation to ordinary Richardson extrapolation

Ordinary Richardson extrapolation assumes an expansion $s_n = L + c_1 h_n + c_2 h_n^2 + \cdots$ and cancels finitely many leading terms. Here $h_n = 4^{-n}$, but two stronger statements hold:

1. the admissible exponents $1, 2, \ldots, d$ are known exactly from the binary level of x;
2. after exponent d, there is no remainder at all.

Thus (9) is simultaneously a geometric Richardson formula, a Lagrange-evaluation formula at zero, a spectral projector for the shift operator, and an exact inverse-Vandermonde row.

14 Conclusion

For a fixed dyadic rational x, the finite-sinc approximants do not merely approach $\mathrm{up}(x)$ with an asymptotic power series. After finitely many initial stages, their values lie exactly in the finite-dimensional span

$$1, 4^{-n}, 16^{-n}, \ldots, 4^{-dn}, \qquad d = \lfloor \ell(x)/2 \rfloor .$$

The mechanism is structural: the binary denominator aligns x with a finite-spline cell, complete Thue–Morse blocks collapse its local degree, and the self-similar omitted tail acts through an even moment operator whose inverse terminates on that polynomial.

The resulting limit extraction is explicit, rational, stable, and compact:

$$\mathrm{up}(x) = \frac{1}{(1/4;1/4)_d} \sum_{k=0}^{d} (-1)^k 4^{-k(k+1)/2} \begin{bmatrix} d \\ k \end{bmatrix}_{1/4} \mathrm{up}_{N+d-k}(x).$$

The same sample window recovers every geometric coefficient and every relevant even derivative, while the finite q-binomial recurrence supplies an exact consistency check. In this setting, q-Pochhammer extrapolation is not a numerical acceleration heuristic; it is an exact arithmetic identity.

A Derivation of the integer-product weights

For $\rho = 1/4$,

$$(\rho;\rho)_s = \prod_{r=1}^{s}(1-4^{-r}) = 4^{-s(s+1)/2}P_s. \tag{102}$$

Substitute (102) into (10):

$$\begin{aligned} w_{d,i} &= (-1)^{d-i}\frac{4^{-(d-i)(d-i+1)/2}}{4^{-i(i+1)/2}P_i\,4^{-(d-i)(d-i+1)/2}P_{d-i}} \\ &= (-1)^{d-i}\frac{4^{i(i+1)/2}}{P_iP_{d-i}}. \end{aligned}$$

This proves (12). It also explains the conspicuous powers

$$1, 4, 64, 4096, 1048576, \ldots = 4^{i(i+1)/2}$$

in the rightmost numerators of the low-order formulas.

B A compact proof of the finite q-binomial identity

For completeness, define

$$(z;\rho)_d = \prod_{j=0}^{d-1}(1-z\rho^j).$$

Induction on d, using the Gaussian Pascal recurrence, gives

$$(z;\rho)_d = \sum_{k=0}^{d}(-1)^k\rho^{k(k-1)/2}\begin{bmatrix} d \\ k \end{bmatrix}_\rho z^k. \tag{103}$$

Taking $z = \rho\mathsf{E}^{-1}$ and multiplying by E^d yields the extraction operator expansion. Taking $z = \mathsf{E}^{-1}$ and multiplying by E^{d+1} yields the recurrence. Taking $z = -\rho$ yields the absolute weight sum in theorem 7.1.

C Reference formulas in one place

For convenient implementation, the main identities are collected here. Let $x \in (-1,1)$ have exact level m, let $d = \lfloor m/2 \rfloor$, $\rho = 1/4$, and choose $N \geq m$.

Finite spline.

$$q_n = \frac{2^{n(n+1)/2}}{n!}\sum_{k=0}^{2^{n+1}-1}\tau_k\left(x+1-2^{-n-1}-k2^{-n}\right)_+^n.$$

Exact geometric law.

$$q_n = \sum_{j=0}^{d} (-1)^j a_j 4^{-j(n+1)} \operatorname{up}^{(2j)}(x) = \operatorname{up}(x) + \sum_{j=1}^{d} C_j(x)\rho^{jn}.$$

Bell–Bernoulli coefficients.

$$\alpha_j = \frac{|B_{2j}|}{2j(2j)!(1-2^{-2j})}, \qquad j a_j = \sum_{r=1}^{j} r\alpha_r a_{j-r}, \qquad a_0 = 1.$$

Limit extraction.

$$\operatorname{up}(x) = \frac{1}{(\rho;\rho)_d} \sum_{k=0}^{d} (-1)^k \rho^{k(k+1)/2} \begin{bmatrix} d \\ k \end{bmatrix}_\rho q_{N+d-k}.$$

Mode extraction.

$$C_s = \rho^{-sN} \frac{\left[\prod_{j\neq s}(\mathsf{E}-\rho^j)q\right]_N}{(-1)^s \rho^{s(s-1)/2+s(d-s)}(\rho;\rho)_s(\rho;\rho)_{d-s}}.$$

Derivative recovery.

$$\operatorname{up}^{(2s)}(x) = \frac{(-1)^s 4^s}{a_s} C_s.$$

Recurrence.

$$\sum_{k=0}^{d+1} (-1)^k \rho^{k(k-1)/2} \begin{bmatrix} d+1 \\ k \end{bmatrix}_\rho q_{n+d+1-k} = 0.$$

References

[1] V. Reshetnikov, *ProveIt: Analysis/FabiusFunction*, especially the finite dyadic sinc product, exact box-spline, Bell–Bernoulli deconvolution, and geometric Richardson developments, `https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction`, repository consulted 2 September 2026.

[2] J. Arias de Reyna, *Arithmetic of the Fabius function*, 2017, `https://arxiv.org/abs/1702.06487`.

[3] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation of the 1982 article, 2017, `https://arxiv.org/abs/1702.05442`.

[4] J. K. Haugland, *Evaluating the Fabius function*, 2016, `https://arxiv.org/abs/1609.07999`.

[5] NIST Digital Library of Mathematical Functions, *Chapter 17: q-Hypergeometric and Related Functions*, especially Section 17.2 on q-Pochhammer symbols, Gaussian coefficients, and the finite q-binomial theorem, `https://dlmf.nist.gov/17.2`.